A Physically Coupled Robotic Titration Laboratory

Newton 1.5 apparatus, local chemistry and evaluation

Gym Anything for Robotics

This document is not a benchmark-release certificate. The evaluation section distinguishes measured component results from outstanding combined-environment gates. **Acceptance result:** the complete 85 mL HTTP/RGB/history path measures 640.87 wall seconds per simulated second and does not meet the ceiling of 400. Component demonstrations below do not override that result.

Abstract

We construct a robotic titration laboratory around a standard Franka Panda, movable hollow glassware, a through-bored rotary stopcock, a free magnetic stir bar and spatial chemical inventories. Newton 1.5 supplies the articulated rigid world and material-point state. A repository-owned SPH extension supplies water pressure, viscosity, capillarity and rigid-boundary reactions; it is not an undocumented native Newton water solver. A separate renderer consumes recorded state without advancing a second physics world. Evaluation concerns environment construction: component fixtures may apply explicitly labelled external forces, but neither an autonomous titration solution nor unavailable physical calibration is required. This dossier records the mechanisms, parameter sources, observed numerical defects, visual evidence, access restrictions and release gates.



Recorded component scene; source and assistance are identified in the visual-evidence section.

Contents

1 Scope and scene construction **3**

1.1 Dimensional sources and design choices . . . 3

1.2 One geometric source . . . 3

1.3 Room and camera . . . 4

2 Coupled mechanics and liquid **4**

2.1 Engine identity and clock . . . 4

2.2 Water properties and numerical resolution . . . 4

2.3 Surface energy and wetting . . . 5

2.4 Solid boundaries and contact . . . 5

2.5 Robot actuator references . . . 6

2.6 Physical pipette filler and air pressure . . . 6

3 Chemical state and optical observations **6**

3.1 Local conservation and equilibrium . . . 6

3.2 Continuous surface and local colour . . . 7

4 Evaluation: what the apparatus actually does **7**

4.1 Scope and provenance . . . 7

4.2 Combined apparatus and liquid . . . 7

4.3 Pipette uptake and physical robot contact . . . 8

4.4 Other bounded physical checks . . . 8

4.5 Measured operating cost . . . 9

4.6 Observations, controls and recovery . . . 9

4.7 Limits and release decision . . . 9

5 Interfaces, task evaluation and reproduction **10**

5.1 One environment object and one clock . . . 10

5.2 Permissions and durable state . . . 10

5.3 Read-only endpoint verifier . . . 10

5.4 Reproducible catalogue and deployment . . . 11

6 Recorded visual evidence **12**

7 Selected primary sources and implementation boundaries **16**

1 Scope and scene construction

The intended task is a robot-operated acid–base titration. Actions move the robot; contact moves apparatus. Liquid transport and local chemical composition determine the visible indicator, rather than a task-phase colour command. Creating a plausible initial scene is distinct from arranging one to solve the task. In particular, camera visibility from a single initial pose is not a completion gate: the policy may look or move. Sensor mounting, projection, freshness and observation permissions are completion gates.

The environment covers continuous liquid optics, hydraulic behaviour, laboratory appearance, usable physical apparatus and unified interfaces.

1.1 Dimensional sources and design choices

Object	Published reference	Reconstruction / implementation
Corning 4980-125 flask	125 mL nominal capacity; 67 mm maximum diameter; 114 mm height	Hollow meridian mesh with 1.5 mm side-wall and 2 mm bottom engineering estimates; 67 g mass reference. The vessel is a free rigid body.
Corning 2103-50 burette	50 mL; 750 mm overall height; 14 mm diameter; 0.1 mL graduations; 0.05 mL class-A tolerance; 2 mm PTFE valve bore	5.25 mm inner column radius and 0.65 mm drawn-tip bore radius are reconstruction estimates. Graduations integrate the model cavity, not a visually fitted scale.
IKA C-MAG MS 4	150 by 260 by 105 mm envelope; 100 mm square plate; 100–1500 rpm; 2.716 kg	Free housing on four feet, physical speed dial, magnetic rotor field and freely moving PTFE-coated bar.
Corning 7100-10 pipette	10 mL class A; approximately 429 mm length	Hollow stem/bulb/tip profile on an open support rack. Published delivery tolerance is not claimed as achieved by the numerical liquid.
Manual pipette filler	Kartell article 198: 40 mm diameter, 100 mm envelope, nominal 10 mL; ABS with HDPE/PA6 parts	Reconstructed pull-plunger/vent aid, not exact product CAD. Hollow barrel, piston, thumb disc, silicone chuck and spring-return vent are physical components with passive prismatic joints.
Franka Panda	Standard source articulation and meshes	Seven arm joints, parallel fingers, actuator targets in SI units, fixed table base and explicit mount/bolts. No idealized task-specific arm is substituted.
Laboratory	Architectural design rather than manufacturer CAD	1.8 by 0.75 m epoxy bench at 0.9 m; sealed floor, cabinets, open room doorway, glazed window, actual ceiling luminaire housings and external camera stand.

Primary apparatus references are the Corning catalogue, the manufacturer product identifiers above, and the selected sources at the end of this dossier. A manufacturer's dimensional specification does not make an estimated inner profile manufacturer-supplied CAD.

1.2 One geometric source

The apparatus builder supplies the dimensional definitions for both visible and collision geometry. Rigid contact uses separately flagged convex wall cells with real hollow interiors. A convex hull of a whole flask would fill its cavity and is not used. The stopcock mesh is a cylinder with a transverse bore; the socket has matching ports. A revolute bearing constrains the five supported degrees of freedom while leaving rotation free with stated friction. There is no sliding invisible seal.

Valve fluid contacts and solid-volume quadrature use analytic cylinder differences with the same radii and bore. Independently tessellating the plug and socket would round their sharp intersections differently and create a false

fluid passage at the mating surface. The analytic solid sign removes that discretization gap. It does not change the valve's dimensions or its rotary degree of freedom. Render and rigid-contact meshes remain tessellated approximations, and the geometry API distinguishes those representations.

Straight meridian edges are subdivided for numerical conditioning, without changing the piecewise-planar wall surface or instrument dimensions. The bore query experiment measures the difference between this triangulation change and an actual shape change. Material textures and small optical bevels are appearance detail; collision dimensions remain explicit in the geometry API.

1.3 Room and camera

The burette is clamped on a stand behind the free flask and stirrer. Reagent bottles and a rinse vessel are distinct free bodies. The pipette rests across two glass-stem saddles and a third cradle beneath its assembled filler. Cabinets, window frames and lighting fixtures provide spatial and material cues without placing arbitrary clutter into the robot work area. Poly Haven textures are pinned shared assets; attribution and hashes accompany the generated scene.

The policy camera is an explicitly mounted external RGB camera, not a sensor implicitly attributed to the Panda. Teacher diagnostic cameras provide closer apparatus views. Rendering uses metre-scale USD, physically refractive glass, path-traced light transport and source-state body poses. The render worker rejects a USD stage containing a physics scene or dynamic rigid-body API.

2 Coupled mechanics and liquid

2.1 Engine identity and clock

The runtime pins `newton==1.5.0` and `warp-lang==1.16.0`. It rejects version mismatch rather than silently using a fallback engine. Newton's CPU MuJoCo rigid backend integrates articulated bodies and contact; the GPU SPH extension advances Newton material-point states. Isaac RTX is a render-only dependency. There is one simulation clock. At each substep, liquid reactions are accumulated onto the rigid bodies, the magnetic model adds its loads, and the existing Newton rigid solver takes exactly one step.

2.2 Water properties and numerical resolution

At 20 degrees Celsius, the model uses density 998.2 kg/m^3, dynamic viscosity 0.001002 Pa s, and surface tension 0.072736 N/m. The source ledger identifies the water-property references, including IAPWS. Each liquid point represents volume $V = \Delta x^3$ and mass $m = \rho V$. Requested initial volumes must be integer multiples of this parcel volume; the implementation rejects an incompatible request instead of rounding it silently.

The pressure discretization is DFSPH-inspired: divergence and density errors are projected separately using local kernel gradients. Both mean and maximum residuals are recorded. Exceeding the declared iteration budget produces an explicit fault, not an unreported approximate step. A previous-pressure initial guess is an optional numerical optimization; it does not relax the convergence criterion. Initial conditions, timestep, kernel support and solver settings are part of each run's evidence record.

The complete reference inventory is 85 mL: 10 mL acid and 15 mL water in the flask, 20 mL base in the burette, and 20 mL in each source bottle. The reference file specifies 0.625 mm parcel spacing (348160 parcels), 125 microsecond steps, 1.25 mm pressure support and 1.875 mm interface-energy support. Both pressure projections use a 0.001 mean residual criterion and a 1000-iteration limit; the maximum residual is also reported. The 0.0032-spacing numerical wall clearance is 2 micrometres. None of these values is reduced when timing the camera/recording runtime.

The accelerated pressure option uses projected-gradient momentum; a nonnegative iterate is evaluated before it can affect the integrated velocity. Positive compression error alone is an insufficient stopping test for that iteration: it can accept excess repulsion. The implementation therefore measures the projected-gradient complementarity residual in density-ratio units. A declared maximum residual can supplement the mean. Runs 065 and 069 use

the 0.001 mean criterion and record, but do not enforce, a separate maximum. An independent double-precision nonnegative least-squares velocity reference checks this operator against that overshoot. This numerical result does not by itself validate drop formation or wetting.

2.3 Surface energy and wetting

The repository surface-energy formulation follows the construction in Probst and Teschner, *Unified Pressure, Surface Tension and Friction for SPH Fluids*, ACM TOG 44(1), doi:10.1145/3708034. Its regularized density-deficit energy uses

$$\psi(C) = \sqrt{\max(C, 0)^2 + 0.05^2} - 0.05, \qquad A_0 = \frac{\pi}{4}\Delta x^2.$$

The implementation differentiates this energy for liquid and solid-boundary quadrature points and returns the opposite solid reactions. It implements the cited energy, not the paper's entire unified implicit solver.

Wetting uses explicit material-angle references: 25 degrees for the clean-glass reference and 108 degrees for PTFE. The latter is supported by McHale et al., J. Phys. Chem. B 102 (1998), 1964–1967. Other surfaces use a declared 80-degree engineering estimate. These are not specimen-specific measurements, and contact-angle hysteresis is not implemented. Physical calibration is unavailable in this digital project; researched values and stated estimates are evaluated through digital component behaviour.

2.4 Solid boundaries and contact

Body-attached quadrature samples only the union of actual solid interiors. Glass cavities remain empty. Rebuilding the local sampling region does not move, delete or respawn liquid. Exact particle/shape contacts add unilateral velocity constraints. For predicted velocity v_0, the local projection solves

$$\min_v \frac{1}{2}\|v - v_0\|^2 \quad \text{subject to} \quad n_j^T v \geq b_j.$$

In three dimensions, enumerating active sets of at most three planes gives the closest feasible velocity. Small Gram systems are solved in double precision; the Newton state and emitted reaction arrays retain their declared types. The opposite impulse is applied to the contacted body. No point is teleported to repair penetration after integration.

A single boundary search box for the full inventory would be unnecessarily large because it spans separate containers. Boundary quadrature therefore uses the union of guarded, current-position boxes for disjoint parcel-index groups. Every parcel is included even after leaving its original vessel. These groups have no effect on liquid interaction or chemical mixing. Overlapping search regions share one body-local quadrature node; they do not double solid volume. The sample lattice, spacing, geometry and coverage-support margin are unchanged.

The active interior-cylinder branch of an analytic valve contact instead has a finite-step constraint. If a is its axis, $P = I - aa^T/(a^T a)$ projects onto the radial plane, r is the authored radius less the declared numerical skin, and d is the starting relative position minus first-order boundary transport, then

$$\|P(d + \Delta t v)\| \leq r.$$

A tangent plane alone admits a radial crossing of order $(\Delta t v_{\text{tangent}})^2/(2r)$, even when its velocity inequality is satisfied. Dykstra projection handles this disk together with the remaining contact half-spaces. Individual projection corrections retain the opposite surface impulses. This uses a locally selected convex CSG branch; it is not a global nonconvex contact solve or evidence of successful valve throughput.

Known hollow meridians use their authored faceted cavity sign; other closed meshes use generalized winding. Valve Boolean fields give exact solid membership but only a conservative distance bound at intersections. Primitive contacts use Newton's native primitive queries. A pre-integration guard detects nonfinite velocity, motion exceeding half the pressure support in one step, or contact residual above 0.1 micrometre. It stops and records failure rather than clipping speed, hiding a parcel or continuing with corrupt state.

2.5 Robot actuator references

Newton 1.5's MuJoCo backend does not implement the model's velocity-limit attribute as a hard joint-speed constraint. The runtime therefore retains the requested joint destinations separately from motor references, which approach them at the manufacturer URDF velocity limits. Actual positions and velocities remain force/contact-driven; they are never clipped to the reference. The requested destinations, references and native motor targets are checkpointed. The finger rate fixture measured a maximum 0.20000151 m/s against the 0.2 m/s URDF reference rate. External forces can still make actual speed differ.

2.6 Physical pipette filler and air pressure

The manual filler is part of the pipette assembly, not a robot attachment. Its plunger and vent move through physical prismatic joints. The reconstructed 20 mm piston bore, 2 mL dead volume, 1 mm vent diameter and 0.5 N seal-friction parameter are explicit engineering estimates; published product envelopes do not provide these interior values. Ambient air is 101325 Pa at 293.15 K, with specific gas constant 287.05 J/(kg K) and heat-capacity ratio 1.4.

The physical piston permits 12 mL swept volume (38.197 mm travel) within the existing barrel. A nominal 10 mL liquid capacity needs additional displacement for expansion of the air cushion: the authored geometry needs 10.446 mL sweep at a 0.34 m hydrostatic lift. Equating piston displacement with nominal liquid capacity would prevent reaching the calibration mark. The 12 mL travel is a reconstructed mechanical allowance, not an automatic liquid-fill target.

Let $\chi(x, X)$ be a smooth pipette-cavity occupancy function, V_i the constant parcel volumes, s actual piston travel, and A its area. Then

$$V_\ell = \sum_i V_i \chi(x_i, X), \quad V_a = V_{\text{cavity}} + V_{\text{dead}} + As - V_\ell, \quad p_a = m_a RT/V_a.$$

Differentiating the gas work including ambient pressure gives

$$F_i = -(p_a - p_0) V_i \nabla\chi, \qquad F_{\text{piston}} = (p_a - p_0) A.$$

The tool and casing receive equal/opposite loads. This supplies the local pressure difference while the existing liquid solver determines flow; no parcels are inserted, transferred, deleted, or assigned an upward velocity. The open-port pressure transition spans the existing kernel diameter. A one-spacing transition loses quadrature traction between particle layers, as stored positions and an independent lattice check show.

Opening the physical vent exchanges gas with ambient using compressible orifice flow and a mass-conserving backward-Euler solve. A geometric liquid coverage test also vents an unsealed pipette tip. The gas is a lumped connected air cushion, not resolved bubbles or arbitrary disconnected air compartments. The assisted fixture places the tip before tick zero, supports the tool with a declared compliant wrench, and applies bounded forces to the actual plunger and vent. It is component evidence, not robot-operated pipetting autonomy.

3 Chemical state and optical observations

3.1 Local conservation and equilibrium

Five species inventories travel with each parcel: analytical acid excess, base excess, sodium, chloride and phenolphthalein. Transport is a conservative pairwise Fickian exchange with a declared 1.5e-9 m^2/s mutual-diffusion estimate. Advection is the physical parcel motion. Numerical diffusion measured in a PhysX particle liquid is not imported as a molecular material constant. Acid/base neutralization consumes equal local amounts. Sodium, chloride and indicator totals are conserved; reacted acid/base are tracked separately.

For net analytical molarity c, equilibrium solves

$$[H^+] - [OH^-] = c, \qquad [H^+][OH^-] = K_w, \qquad K_w = 10^{-14.17}.$$

The alkaline branch is evaluated without subtractive cancellation. The coloured indicator concentration is

$$c_{\text{pink}} = \frac{c_{\text{indicator}}}{1 + 10^{pK_a - pH}}, \qquad pK_a = 9.7.$$

The 25-degree indicator reference is an explicitly documented approximation near 20 degrees. High-pH fading and a full activity-coefficient model are not implied by this equilibrium equation.

3.2 Continuous surface and local colour

The renderer reconstructs a fixed 0.5 Wendland volume-fraction isosurface. Sparse occupied bricks avoid sampling the large empty volume between flask and burette. Seams are merged without changing particle positions. Reconstructed mesh volume is reported separately from material volume and is not rescaled to force agreement. Unresolved isolated parcels remain visible as counted equal-volume glyphs; they are not silently removed.

Indicator concentration is interpolated locally onto the reconstructed surface. An absorption material uses the natural-log Beer–Lambert coefficient

$$I/I_0 = \exp(-\epsilon c_{\text{pink}} L), \quad \epsilon = 2.935 \times 10^6 \ (\text{M m})^{-1}.$$

This coefficient comes from Mueschke et al., UCRL-JRNL-237116, printed p.10, Eq.2.14. No factor of $\ln 10$ is inserted because the source uses natural logarithms. The green absorption band is reduced to an RGB channel approximation. Boundary-sampled heterogeneous absorption is not a claim of fully resolved spectral volumetric optics.

4 Evaluation: what the apparatus actually does

4.1 Scope and provenance

These experiments evaluate environment construction, not a robot policy. Assisted fixtures may initialize a pose or charge and apply a declared, force-limited support or control input. Liquid motion, contact, valve flow, stirring and local colour are simulated, not prescribed outcomes. Every listed run retains its effective configuration, timestamps and result. Rendering recorded states takes no physics steps. Independent runs and checkpoint restorations are never joined into one continuous video.

4.2 Combined apparatus and liquid

Run 169 contains the complete 85 mL reference charge and standard Panda. It advances 24000 consecutive steps at 125 microseconds (3 simulated seconds). The flask, magnetic bar and reagent vessels remain movable. Assistance applies at most 0.02 N m to the actual stopcock joint and 0.03 N m to the stirrer dial; neither assistance prescribes liquid speed or stir-bar motion.

The angle convention matters: the bore is vertical and *open* at 0 rad; it is sideways and *closed* at $\pi/2$ rad. Run 145 labels these endpoints the other way round. Run 169 uses a closed/part-open/open/closed schedule.

In 241 recorded frames, no parcel centre lies inside either the analytic PTFE plug or glass socket. No burette material crosses the outlet during the initial closed hold. First outlet crossing occurs at tick 3100 after the opening command at tick 1600. Delivery rises through partial and full opening. After the reclosure command at tick 14400, material already in the outlet finishes draining: the below-tip inventory reaches 1.875732 mL at tick 16300 and remains exactly constant through tick 24000. All of it is in the flask at the final sample. Total represented liquid remains 85 mL, with no final uncontained material or bench spill. The geometric audit is every 100 ticks, not every substep.

The free bar makes 13.513672 revolutions relative to the moving stirrer housing. Its maximum axial vertical component is 0.00225, so planar rotation is a useful measurement here. Chloride concentration coefficient of variation on the *original* flask material falls from 1.224745 to 1.189530. Tracking that fixed cohort prevents an apparent improvement merely from adding clear titrant. This demonstrates local mixing; three seconds do not establish a uniform, thirty-second endpoint.

Video 175 shows that uninterrupted run from tick 0 through 24000. All 181 encoded frames decode, and eight actual frames spanning closure, opening, delivery and reclosure were opened along with four final camera views. The video is labelled 2x slow motion and records no RGB-readback recovery. At tick 0 the initialized flat liquid surface appears darker than later frames; the cause of that visual difference is not established here. The original frame remains in the video rather than being replaced or omitted.

4.3 Pipette uptake and physical robot contact

Run 119 records one uninterrupted 2.6-second uptake/hold/release cycle. The tip is initially immersed, and a declared compliant support holds the tool. Forces of at most 3 N operate the real plunger and vent. With 8 mL in the source, 0.302002 mL is geometrically inside the pipette before venting. After drainage, 0.010742 mL remains. All 8 mL remain represented. The pressure and uptake curves are measured from particle positions and gas state, not a commanded transfer.

That is a partial-fill demonstration. A 10 mL piston sweep cannot lift 10 mL of liquid: trapped air must also expand. The minimum hydrostatic sweep for the reconstructed instrument is 10.445513 mL. Its 12 mL maximum sweep fits inside the barrel. The larger trial 157, commanding a 10.7 mL piston sweep over 5 s, stopped at tick 15538: one candidate parcel exceeded half-support travel. This was an explicit rejection before integration, not a successful run or a nonfinite state.

Run 172 keeps the 20 mL source, 10.7 mL piston reference and 3 N force limit, but ramps the reference over 10 s. It completes 124800 consecutive steps (15.6 s), including a 0.5 s hold and 5 s of vented drainage. Peak actual uptake is 9.765137 mL. At the end, 3.707520 mL remains in the pipette and 16.292480 mL is in the source bottle. All 20 mL remain represented; an independent read-only audit finds no material outside their circular body-local cavity profiles in 1249 sampled states and agrees with recorded uptake to floating-point roundoff. This is near-nominal uptake and demonstrated drainage, not complete emptying within 5 s or reproduction of the manufacturer's delivery tolerance.

The piston reference is held during the hold phase, not the liquid surface: uptake continues from 9.472168 to 9.761719 mL while the pressure relaxes. Vent operation brings final gauge pressure to -3.87 Pa. The complete fixture takes 2358.779 wall seconds /15.6 simulated seconds =151.204 wall/simulation, without online RGB/history. It is not substituted for the full laboratory performance benchmark. No physical value was changed to obtain this slower-fill result.

Video 180 renders the complete 15.6-second run at 1x speed. All 469 frames decode; nine actual frames from tick zero, filling, the reference hold and drainage were opened. The full tool stays within the documentary view, including the moving piston thumb disc. The clear column is subtle at this wide scale, so quantitative uptake comes from the recorded-state audit. The renderer reports no RGB-readback recovery. No checkpoint, separate trial or interpolated physical state is substituted into the recording.

Run 124 tests the assembled filler between Panda fingers, without a robot weld or post-initialization support. Recorded contact has four points per finger, approximately 19.12 N per finger, and final penetration 8.04 micrometres. The tool remains between the fingers after settling. Initial tool motion includes gravity-driven arm settling; it is not all interpreted as grasp slip. Render 147 is a continuous 61-frame, 2-second video, with initial, intermediate and final decoded images inspected. This is contact/retention evidence, not autonomous pipetting.

4.4 Other bounded physical checks

An independently free 10.048 mL flask fixture measured mean downward liquid load 0.0983546 N against its material weight 0.0983599 N, with no recorded parcel centre inside glass or below the floor at 0.75 s. Its sparse liquid surface enclosed 9.876978 mL, approximately 1.70 percent below the material volume. The renderer reports that bias rather than rescaling the visible liquid.

The torque-driven 45 mL combined run 099 delivered 2.317383 mL in 2 s with zero valve intrusions in 161 saved frames. It is separate evidence from the 85 mL run, and its 382.87 wall/simulation ratio excludes online RGB and HTTP histories. It must not be used to claim full-runtime performance acceptance.

Independent local checks accompany the apparatus measurements: force versus energy derivative, equal/opposite reactions, constrained contact solutions, species conservation and typed checkpoint round trips. They identify

specific implementation mistakes; their count does not establish visual realism.

4.5 Measured operating cost

The acceptance ceiling is a ratio **below 400**. The operational benchmark uses all 85 mL, 348160 parcels, one environment, the 32-sample 1280-by-720 RGB camera, every-substep rigid/pressure history and full liquid trace arrays every 20 ticks. After 100 warm-up ticks, four consecutive 800-tick HTTP hold commands advance 0.4 s. Startup is reported separately.

Run 156 takes 256.347483 wall seconds for those 0.4 simulated seconds: **640.868707 wall seconds per simulated second**. No action or trace capture failed, but performance **does not pass 400**. Lossless binary recording, conservative boundary-query invalidation and spatially coherent GPU work scheduling reduce unnecessary work without changing material values, time resolution or observation cadence. The cost plot separates liquid/chemistry, rigid mechanics, durable recording, RGB and other overhead. Lower sample counts or a reduced liquid charge are not substituted for this measurement. No aggregate parallel throughput is claimed.

A comparison run (170) changes only the previous-pressure initial guess fraction from 0.5 to 1.0, retaining force equations and convergence criteria. It measures 603.575687 wall/simulation. This also fails 400, and a single comparison does not establish a statistically reliable speedup. The reference remains at 0.5. The candidate is not a performance-qualified replacement.

Spatial scheduling was compared on identical saved arrays without integrating physics. Particle-local capillary energy and force fields remain bit-identical; opposite solid reaction differs by at most 3.64×10^{-12} N due to atomic summation order. The neighbour-cache schedule is left unchanged because it did not get faster. These are implementation checks, not material fitting.

4.6 Observations, controls and recovery

Actual authenticated HTTP fixtures exercise actions, role isolation, idempotency, history and checkpoint operations. The manual dashboard has been operated in Chrome, including typing isolation and rejected stale requests. Checkpoint restoration is labelled discontinuous and is not video evidence. An injected storage failure is rejected explicitly rather than silently reporting an unrecorded action as successful.

Current-source checkpoint fixture 174 advances a labelled 1 mL component, serializes its complete state, advances again, then restores. Every saved array and scalar matches exactly after restoration, including liquid, species, pressure history and the air circuit. It also restores the initial state and advances once. This verifies data round-trip/recovery, not bit-deterministic future replay, full-inventory performance or video continuity.

An RTX readback failure can stop a video at a black frame. The renderer therefore performs a synchronous capture and a bounded render-product recreation if readback remains invalid. This never resets the simulator or advances physics. A moving-geometry camera fixture measured at most 0.5772 pixels discrepancy from independent pinhole projection across four alternating poses. That establishes current-frame geometry in the fixture; the liquid video has its own inspection requirement. Video 158 completed all 157 frames with no readback retries or render-product recovery. Every frame decoded; seven actual frames spanning the full recording were opened. It presents the continuous 2.6-second fixture 119 at 2x slow motion. The close-up shows the immersed tip and changing source surface; the bottle label hides the upper column, so it is not used alone to measure uptake.

4.7 Limits and release decision

Physical-world calibration is unavailable and is not a completion prerequisite. Dimensions, materials and estimates have the research basis recorded in this dossier and its source ledgers. The relevant numerical limitations are finite liquid resolution, unresolved thin films/small droplets, one connected lumped air cushion, no wetting hysteresis, a dilute ideal-solution approximation and RGB rather than spectral absorption. None is hidden by a global transfer, colour command or particle deletion.

Full-runtime performance below 400 remains unmet. The faster five-second pipette-reference ramp is also not a validated operating recipe: its numerical travel guard stopped the run. The successful slower fixture is not used to

conceal that failure or to claim a complete operating envelope. Default benchmark qualification remains false. An available API or successful PDF build is not reported as physical completion.

5 Interfaces, task evaluation and reproduction

5.1 One environment object and one clock

The public identifier is `titration`, with alias `titrate` and the Panda robot. An explicit dry-apparatus choice or liquid numerical configuration is currently required; `gar.create("titration")` without that choice rejects before starting a simulator. Registry presence does not certify the liquid benchmark.

Actions specify named Panda joint targets in radians, finger targets in metres, and an exact integer physics-tick count. A hold action uses an empty target object. No HTTP action teleports an object, welds a grasp, or directly turns the valve. Assisted component scripts have a separate, explicitly labelled wrench interface that is not a restricted policy action.

The owner manual dashboard uses the same serialized HTTP service. It shows the current physics tick, measured joints and a free-camera geometry preview. Actual RGB is produced from the same simulator state by the renderer. Operator requests carry the expected generation and tick, rejecting stale moves with HTTP 409. Idempotency keys prevent an uncertain reply from causing a repeated action. Typing in an input is not a robot movement command.

5.2 Permissions and durable state

The owner key exposes geometry, current privileged state, checkpoint operations, variation changes and command/substep history. The restricted key provides documented robot joint controls, joint proprioception and the configured RGB camera. Chemical inventories, contact details, geometry queries, checkpoint operations and variation starts are privileged. Simulator-only endpoint reward and success termination are also withheld from restricted actions.

Each physical step records robot/object poses, velocities, targets, rigid contacts and event changes. Liquid arrays use an explicitly reported stride; the default is 20 ticks, not a claim of every-substep full particle recording. Checkpoint data include liquid positions and velocities, local chemistry, pressure history, moving boundary sampling, mechanical state, drive phase and endpoint timer. Restore validates schema, numerical configuration and source identity. Reset/restore changes the episode generation and breaks continuity. Finite numerical arrays are copied before a command and stored losslessly, without expanding every particle coordinate into a JSON number in the stepping path. Titration selects Zstandard compression; privileged retrieval expands the same public JSON structure. Legacy records remain readable. Retention budgets and eviction metadata remain explicit rather than dropping sampling resolution silently.

5.3 Read-only endpoint verifier

Evaluation consumes every consecutive physics tick, beginning at zero. The indicator-hold timer is therefore independent of HTTP polling frequency. It starts when at least 90 percent of indicator-bearing flask parcels have ionized fraction above 0.3; it resets below 85 percent. Thirty simulated seconds of sustained colour are required. The dose tolerance is 0.35 mL. It does not claim class-A metrological accuracy. The spill budget is 0.05 mL.

Sodium in the actual moving flask measures retained analytical base, including neutralized material. Chloride verifies that the original acid was not discarded to obtain an easier endpoint. Sodium, chloride, indicator, and each acid/base inventory plus reacted amount must remain conserved. These measurements are read-only; they do not recolour or reposition liquid. A passing task result would not itself certify hydraulic accuracy or environment quality. Liquid inside an immersed pipette is assigned to its lumen, not counted again as source-vessel contents or as a delivered dose. The actual transported inventory, not an assigned transfer counter, determines this distinction.

5.4 Reproducible catalogue and deployment

The defined catalogue contains 1100 combinations: 25 bench apparatus positions, 11 initial aliquots and four starting burette charges. ID zero is the reference 10 mL acid plus 15 mL water, with 20 mL in the burette. Catalogue definitions are deterministic and preserve requested fluid resolution; not all 1100 combinations have undergone physical qualification. Authenticated `/v1/variations/select` previews without mutation, and `/v1/variations/start` replaces the episode only after a candidate is built and checked. Exact unsupported constraints are rejected rather than rounded.

The renderer installation is selected explicitly; API secrets stay in the local run directory. Assisted valve, mixing, pipette and grasp recordings establish the specific component behaviours reported here. They are not a complete operating envelope or a robot solution. The complete HTTP runtime still exceeds the performance ceiling of 400 wall seconds per simulated second.

6 Recorded visual evidence



Figure 1: The standard Panda and reconstructed laboratory after full-charge fixture 169 (render 175). All 85 mL are represented. Cabinets, luminaires, camera stand, mounts, hollow glassware and the assembled pipette filler are authored at physical scale. The robot is not executing an autonomous policy.

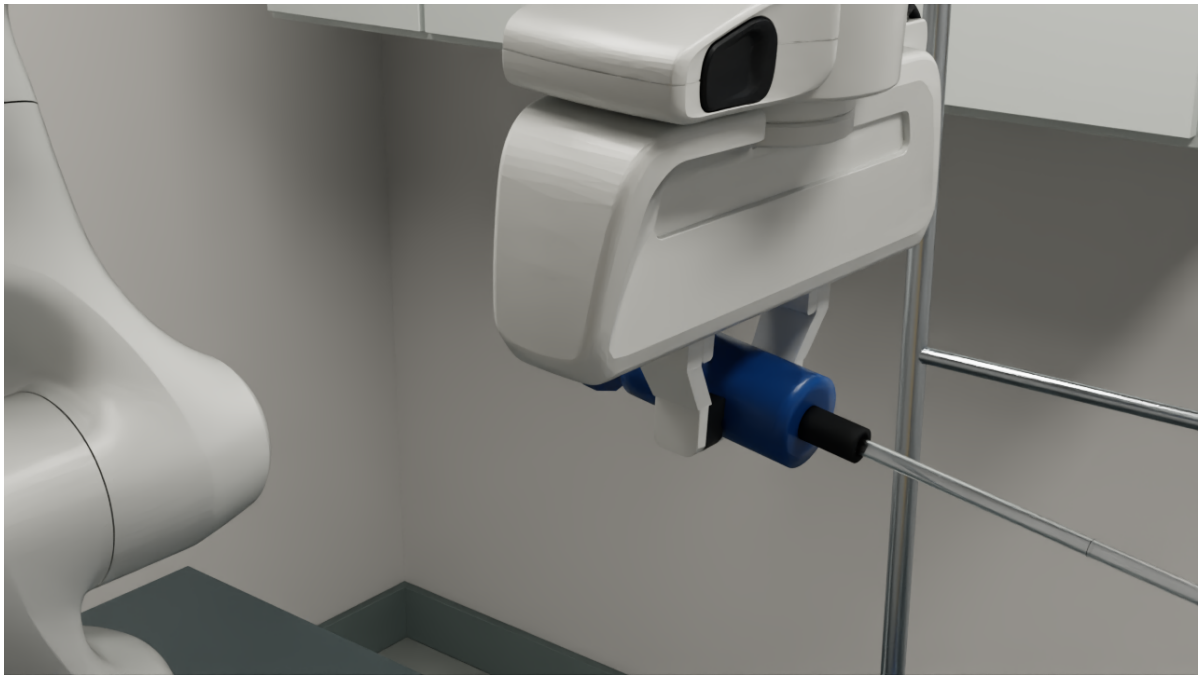


Figure 2: The physical filler barrel retained between the Panda fingers. The companion 61-frame video shows the continuous 2-second dry contact fixture. The tool is not welded to the robot and no support acts on it after initialization.

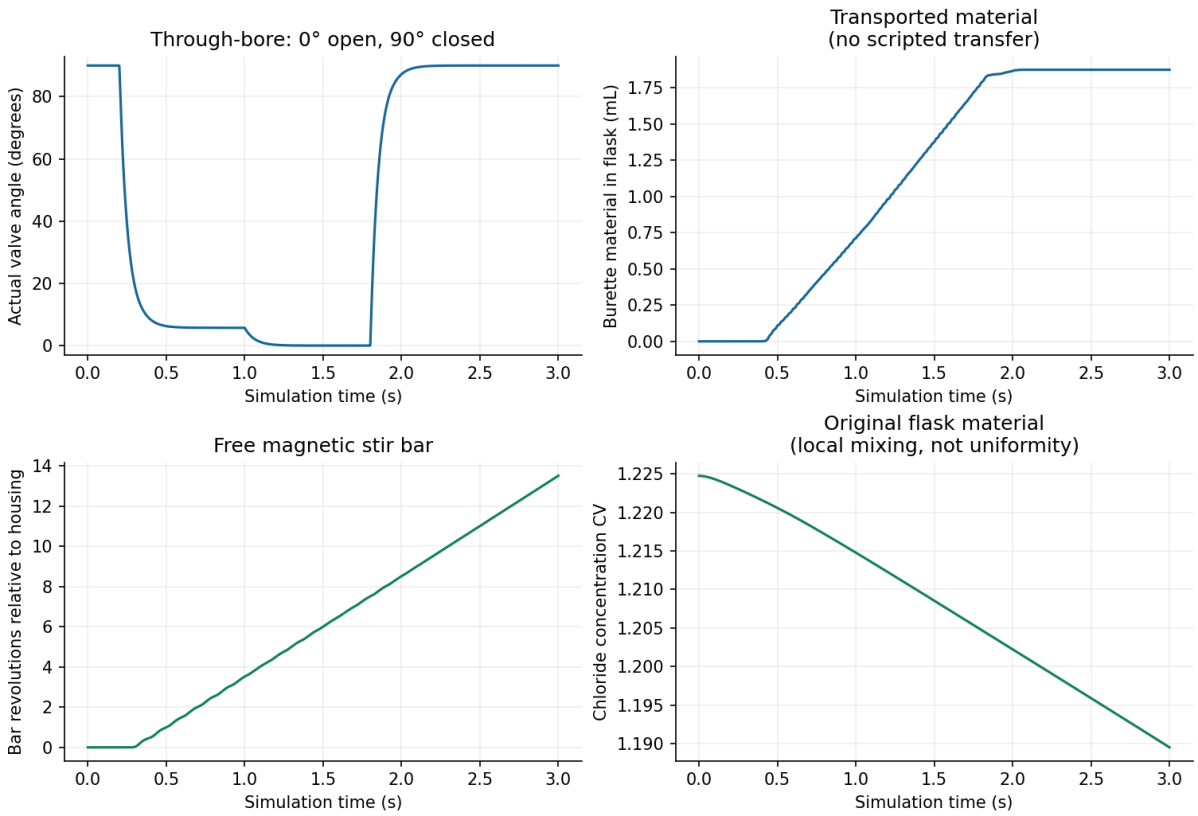


Figure 3: Complete 85 mL coupled fixture 169, plotted directly from recorded states. The sequence is closed, partly open, fully open, then closed: 0 degrees is open and 90 degrees closed. After residual outlet drainage, delivery stays at 1.875732 mL. The free magnetic bar rotates and conserved-tracer variation decreases, without claiming uniform mixing.

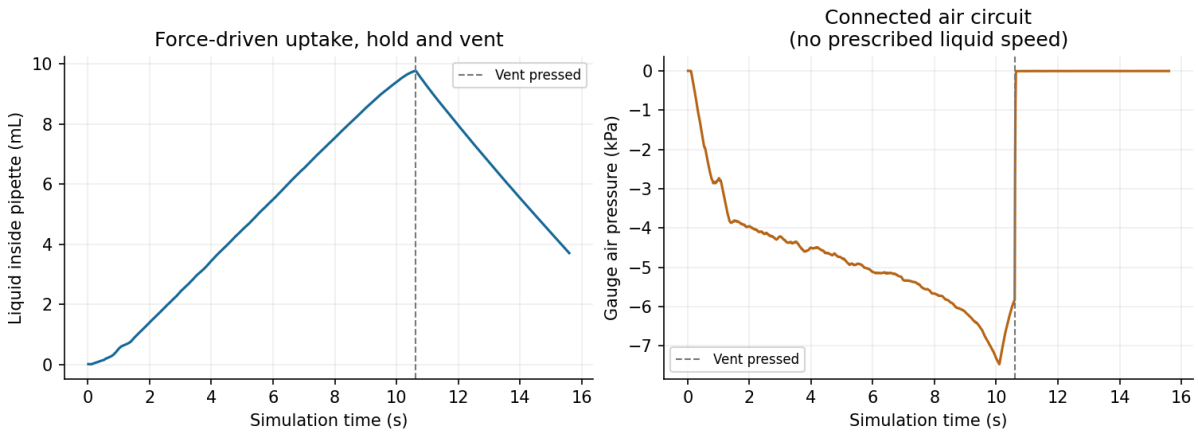


Figure 4: Uninterrupted near-nominal fill, piston-reference hold and vented drainage in fixture 172. Peak uptake is 9.765 mL; 3.708 mL remains after five seconds of venting. All 20 mL remain represented. The dashed line marks recorded vent assistance, sampled every 100 ticks. No desired liquid volume or trajectory is imposed.

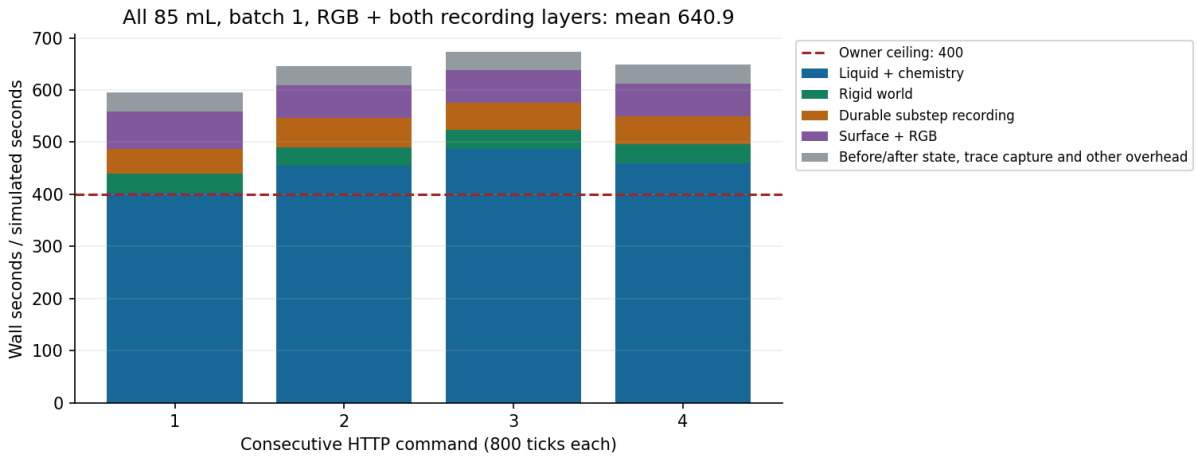


Figure 5: Actual full-inventory HTTP benchmark 156, including live RGB and both history layers. All four commands use 800 ticks at 125 microseconds, after 100 warm-up ticks. The mean 640.87 wall/simulation ratio fails the 400 ceiling. Components are nonoverlapping measured stages; the grey remainder includes snapshot capture and other overhead.

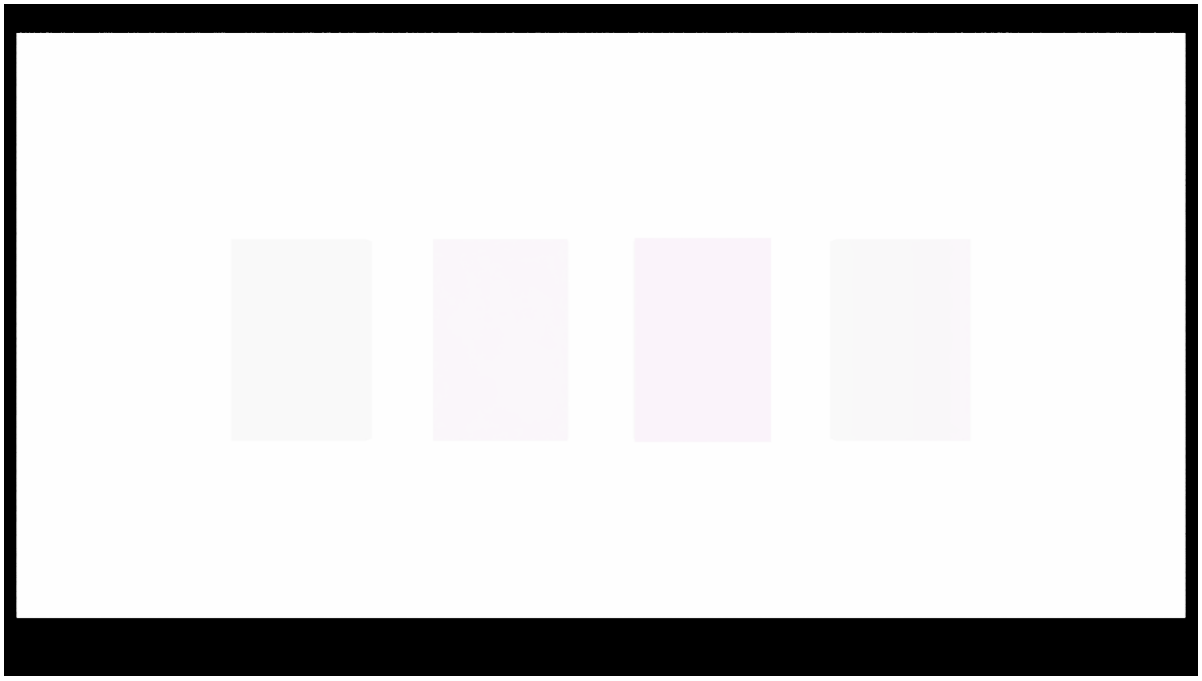


Figure 6: Controlled optical specimens: clear water at 10 mm depth; 10 micromolar pink indicator at 10 mm; the same at 20 mm; and a spatial 0–10 micromolar gradient. These direct RTX renders check depth and concentration-dependent absorption. They are not spectral-camera measurements or a robot task result.

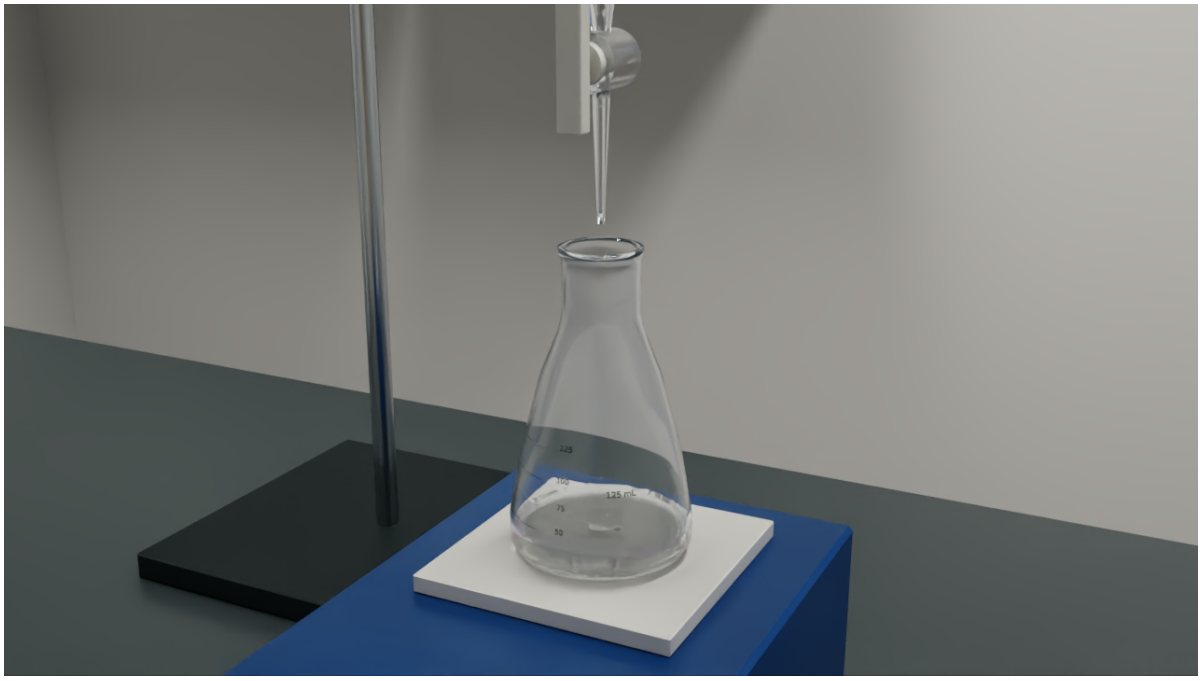


Figure 7: Actual final outlet and receiving-flask view from the continuous valve cycle 169. The outlet has stopped after reclosure and residual drainage. Liquid amount and chemistry come from the recorded material state; the flask is still acidic, so no global pink endpoint colour is applied.

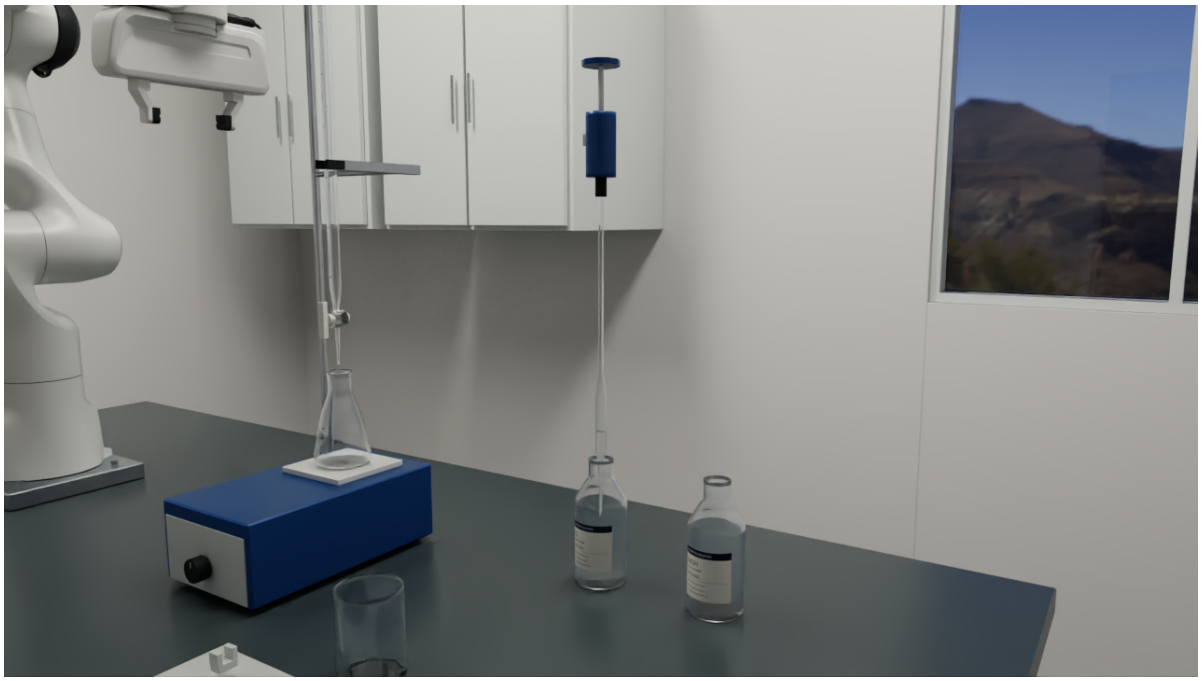


Figure 8: The entire supported physical pipette/filler after fixture 172, with 3.708 mL still in the pipette after five seconds of venting. The documentary camera exposes the plunger, tool and source without changing the policy camera or apparatus. The continuous companion video includes the earlier 9.765 mL peak; this still is the final drainage state, not the peak.

7 Selected primary sources and implementation boundaries

This list identifies the sources used by the reconstruction. Published dimensions and material values are distinguished from estimated interiors and from numerical discretization choices. No physical calibration is required.

1. Newton 1.5.0 release and the pinned installed implementation. Newton supplies rigid mechanics and material-point state; the repository supplies the disclosed SPH/chemistry extension. Availability of a native MPM example is not evidence of native SPH.

2. IAPWS R1-76(2014), surface tension of ordinary water. Temperature-dependent SI surface tension, evaluated at 293.15 K. The coefficient is not replaced by a scene-tuned cohesion strength.

3. IAPWS water-ionization release. Water autoionization near neutrality; the model uses the stated 20-degree approximation $pK_w = 14.17$ rather than ignoring water ions.

4. Probst and Teschner, Unified Pressure, Surface Tension and Friction for SPH Fluids, ACM TOG 44(1). Interface-energy construction, its derivative and published normalization. The complete implicit method in that paper is not claimed.

5. Corning PYREX 2103-50 burette. 50 mL nominal capacity, 14 mm outside diameter, approximately 750 mm height, 0.1 mL scale and 2 mm PTFE bore. Inner column and tip profiles are estimates.

6. Corning PYREX 7100-10 pipette. 10 mL nominal capacity and approximately 429 mm length. Geometric capacity does not claim the manufacturer's delivered-volume tolerance.

7. IKA C-MAG MS 4 technical data. Envelope, mass, plate size and speed range. Internal magnetic moment and drive parameters are disclosed reconstruction estimates.

8. Kartell article 198 plunger filler and Bel-Art Pipette Pump III instructions. Support a physical air-displacement filling aid with a separate release control. The reconstructed piston, seal, chuck and vent are not exact product CAD.

9. NASA Glenn compressible mass-flow derivation and EP0562358B2 air-cushion derivation. Gas mass, pressure and volume are coupled to the piston and actual liquid occupancy; the vent model includes the sonic-flow limit.

10. McHale et al., Evaporation and the wetting of a low-energy solid surface, J. Phys. Chem. B 102 (1998). The reported 108-degree Teflon/water reference supports the chosen PTFE condition, not universal wetting hysteresis.

11. Mueschke et al., Measurements of Molecular Mixing in a High Schmidt Number Rayleigh–Taylor Mixing Layer, UCRL-JRNL-237116, Eq. 2.14. Natural-log green-band attenuation coefficient; converted from centimetres to metres without inserting a factor of $\ln 10$.

12. Warp spatial hash-grid thread mapping. The installed 1.16 implementation was also inspected. Spatial scheduling changes work order, not particle identity, force laws or physical sampling.